ModelIt!

Teacher's Guide

Why Do Some Animals Survive Winter?

The Hidden Science of Insulation, Metabolism, and Staying Alive

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-LS1-1, 4-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-LS1-1, 4-LS1-2

Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a five-component model showing how temperature and food availability interact with body insulation and metabolic rate to determine an animal's survival chance.
Disciplinary Core Idea	**LS1.A Structure and Function** — Plants and animals have both internal and external structures that serve various functions in growth, survival, behavior, and reproduction. Body insulation is an internal structure that functions to maintain body temperature.
Crosscutting Concept	**Systems and System Models** — Students explore an animal survival system with multiple interacting components, competing influences, and emergent outcomes that depend on the balance between energy supply and energy demand.

CAST Testing Alignment

MULTIPLE CHOICE:

An arctic fox has very thick fur while a desert fox has thin fur. Which best explains why the arctic fox's thick fur helps it survive?

CONSTRUCTED RESPONSE:

Using what you know about insulation, metabolic rate, and survival, explain why some animals hibernate during winter instead of staying active. Use the words 'metabolic rate,' 'insulation,' and 'energy' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every winter, animals face a life-or-death challenge: stay warm enough to survive while having enough energy to fuel their bodies. The strategies animals use — from thick fur to hibernation to migration — are the result of millions of years of evolution solving this energy balance problem.

The Cold Challenge

Warm-blooded animals (mammals and birds) must maintain a constant internal body temperature regardless of the outside temperature. A human's body stays at about 98.6 degrees F (37 degrees C) whether it is 100 degrees F or -20 degrees F outside. This is called thermoregulation. In cold weather, heat constantly flows from the warm body to the cold environment. The bigger the temperature difference, the faster heat escapes. An animal must generate enough internal heat to replace what is lost, or its body temperature drops and it dies.

Insulation: The First Line of Defense

Insulation is any material that slows down heat transfer. In animals, insulation takes several forms: fur (mammals), feathers (birds), and blubber or fat layers (marine mammals, bears). Arctic animals have evolved remarkable insulation. Polar bear fur has two layers — a dense underfur and longer guard hairs that trap air close to the skin. Penguin feathers are so dense that they create a nearly waterproof layer that traps warm air. A walrus's blubber can be 6 inches thick, creating an insulating barrier between warm blood and freezing ocean water.

Metabolic Rate: The Energy Engine

Metabolism is the process of converting food into energy. Metabolic rate measures how fast this happens. In cold conditions, an animal's metabolic rate increases because the body needs to generate more heat to maintain its temperature. This means the animal needs MORE food in winter — exactly when food is often scarce. This creates a dangerous energy equation: the colder it is, the more energy the body needs, but the harder food is to find.

Survival Strategies: Insulate, Migrate, or Hibernate

Animals have evolved three main strategies for surviving winter. First, INSULATE: grow thick fur, feathers, or fat to reduce heat loss and keep metabolic rate manageable (arctic fox, polar bear, snowy owl). Second, MIGRATE: travel to warmer areas where food is plentiful and heat loss is lower (monarch butterfly, arctic tern, caribou). Third, HIBERNATE: dramatically reduce metabolic rate to near-zero, living off stored fat, and barely using any energy for months (ground squirrels, some bears, wood frogs). Each strategy solves the same problem differently — balancing energy supply with energy demand.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Temperature and Food Availability as external components (environmental conditions the animal cannot control) and Body Insulation, Metabolic Rate, and Survival Chance as internal components that respond to conditions.

E **Establish Relationships**

Students determine that lower Temperature increases Metabolic Rate (negative), Food Availability increases Survival (positive), Body Insulation decreases Metabolic Rate (negative — insulation reduces energy needs), and high Metabolic Rate decreases Survival (negative — burning too much energy without food is fatal).

V **Visualize & Model**

Students build a five-component model showing competing influences on survival: temperature drives up metabolic rate while insulation drives it down, and food availability must match energy demand for the animal to survive.

E **Evaluate Outcomes**

Students run four scenarios to observe how changing temperature and food availability creates different survival outcomes, and how insulation acts as a buffer that shifts the survival threshold.

R **Revise & Extend**

Students add body size or group behavior to explore why arctic animals tend to be larger (less surface area per volume = less heat loss) and why penguins huddle in groups.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, at the North Pole, it is negative 30 degrees. A polar bear is walking around in a T-shirt of fur. How is that possible?"*

Watch For

Show images of arctic animals in extreme cold. Ask: How do THEY survive when WE would freeze in minutes?

Slide 2: Learning Objectives

3 min

Say: *"Today we are building a model of SURVIVAL — figuring out what makes the difference between an animal that thrives in winter and one that does not."*

Watch For

Have students read objectives aloud. Pre-teach metabolic rate using the analogy of a car engine running faster in cold weather.

Slide 3: Big Question

3 min

Say: *"Your body temperature is 98.6 degrees. An arctic fox keeps the same temperature in minus 50 degree weather. How?"*

Watch For

Have students turn and talk: What do you think arctic animals have that we do not?

Slide 4: LEVER Framework

2 min

Say: *"Our model today has two things the ENVIRONMENT controls and three things that happen inside the animal's body."*

Watch For

Preview five components. Explain the concept of external = environment and internal = the animal's body response.

Slide 5: Activity 1: Components

8 min

Say: *"Can the animal control the temperature? Can it control how much food is in the forest? These are things that happen TO it."*

Watch For

Guide sorting. Temperature and Food Availability are external (the environment controls them). Body Insulation, Metabolic Rate, and Survival Chance are internal.

Slide 6: Activity 2: Connections

10 min

Say: *"Here is the key — when temperature DROPS, metabolic rate goes UP. That is a NEGATIVE relationship. Why is high metabolic rate dangerous?"*

Watch For

Help students map relationships. Focus on the negative connections and why they create a survival challenge.

Slide 7: Activity 3: Simulation

12 min

Say: *"Let us send our animal into the worst winter possible and see what happens. Then let us give it thick fur and see if it survives."*

Watch For

Run mild winter first, then harsh winter (students see survival drop), then harsh winter with good insulation (survival recovers).

Slide 8: Discoveries

5 min

Say: *"Now we know WHY arctic animals have thick fur — it is not just about being warm, it is about keeping metabolic rate LOW so they do not burn through their energy."*

Watch For

Compare three strategies: insulate, migrate, hibernate. Which solves the energy equation differently?

Slide 9: STEM Challenge

12 min

Say: *"You are wildlife rescue engineers. Design a shelter that keeps this warm water from losing heat — just like fur*

keeps an animal warm!"

Watch For

Distribute materials (cups, warm water, thermometers, cotton, bubble wrap, foil, newspaper). Groups test insulation and record temperature every 2 minutes.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Temperature, Food Availability
- **INTERNAL:** Body Insulation, Metabolic Rate, Survival Chance

Relationships

- **Temperature to Metabolic Rate:** NEGATIVE (-) — When temperature DROPS (gets colder), metabolic rate INCREASES — the animal's body works harder to generate heat. In freezing conditions, the body burns energy much faster than in mild weather. This is a negative relationship because the variables move in opposite directions.
- **Food Availability to Survival Chance:** POSITIVE (+) — More food means more energy available to fuel the body's metabolic needs. When food is plentiful, the animal can sustain a high metabolic rate and survive. When food is scarce, the animal runs out of energy and survival drops.
- **Body Insulation to Metabolic Rate:** NEGATIVE (-) — Better insulation means less heat escapes from the body, so the body does not need to burn energy as fast to stay warm. A well-insulated animal has a LOWER metabolic rate in the same cold conditions compared to a poorly insulated animal.
- **Metabolic Rate to Survival Chance:** NEGATIVE (-) — A HIGHER metabolic rate means the animal is burning through its energy reserves faster. If the animal cannot replace that energy through food, high metabolic rate leads to starvation and death. This is why uncontrolled high metabolic rate is dangerous.
- **Food Availability to Body Insulation:** POSITIVE (+) — More food allows the animal to build up fat stores, which provide both energy reserves and physical insulation. Well-fed animals develop thicker fat layers before winter, improving their insulation. Starving animals lose fat and become poorly insulated.

Key Discoveries

- When temperature drops, metabolic rate increases — the animal's body works harder to generate heat, burning more energy
- Better body insulation reduces metabolic rate because less heat escapes, so the body does not have to work as hard to stay warm
- High metabolic rate without enough food is a death sentence — the animal burns through its energy reserves and cannot survive
- Animals survive winter through a combination of insulation (keeping heat in), food stores (energy supply), and in some cases hibernation (reducing energy demand)

Multiple Choice Post-Assessment Answer Key

1. An arctic fox has thick fur and a layer of fat under its skin. How do these features help it survive freezing temperatures?

Correct: B) They provide insulation that slows down heat loss, so the fox's body does not have to burn energy as fast to stay warm

That is right! Fur and fat act as insulation, keeping body heat from escaping. This means the fox's metabolic rate does not have to spike as high, saving precious energy during winter.

2. In the model, Temperature and Food Availability are external components. Why are these external rather than internal?

Correct: B) Because they are environmental conditions the animal cannot control, and the other components respond to them

Correct! External components are the inputs or conditions that drive the system. An animal cannot control the weather or how much food exists in the environment. The internal components (Body Insulation, Metabolic Rate, Survival Chance) respond to these conditions.

3. The model shows that when Temperature drops, Metabolic Rate increases. This is a NEGATIVE relationship. Why does the animal's body burn energy faster in the cold?

Correct: B) Because the body must generate more heat to maintain a safe internal temperature when it is cold outside

Exactly! Warm-blooded animals must keep their body temperature constant. When it is freezing outside, the body works harder (burns more energy) to produce heat and stay warm. This is why cold weather demands more food energy.

4. A student runs the model with Temperature at 10% and Food Availability at 20%. Survival Chance drops very low. Which best explains why?

Correct: B) Cold temperatures raise the metabolic rate, but there is not enough food to fuel that high energy demand, so the animal runs out of energy

Right! Extreme cold forces the metabolic rate up, meaning the body burns energy very fast. But with very little food, the animal cannot replace that energy. It burns through its reserves and survival chance drops. This is the deadly combination.

5. The model shows that Body Insulation has a NEGATIVE relationship with Metabolic Rate. How does better insulation help an animal survive?

Correct: B) Better insulation traps body heat inside, so the body does not need to burn as much energy to stay warm, leaving more energy available for survival

That is right! When insulation keeps heat from escaping, the body does not have to work as hard (lower metabolic rate) to maintain its temperature. This means the animal uses less energy and can survive longer on less food.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Fur and coats create heat"	Insulation does NOT create heat — it only SLOWS the loss of heat that the body is already producing. A coat does not warm you up; it keeps YOUR body heat from escaping into the cold air. If you put a coat on a snowman, it would actually stay frozen LONGER because the coat would keep cold air from warming the snow.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Animals do not feel cold because they have fur"	Arctic animals DO feel cold, and their bodies DO respond to it. Their metabolic rate increases in cold weather just like ours does. The difference is that their insulation is so effective that the metabolic increase is manageable. A polar bear in extreme cold still burns more energy than a polar bear in mild weather — the fur just makes the difference smaller.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Hibernation is just sleeping"	Hibernation is NOT sleep — it is a dramatic physiological shutdown. A hibernating ground squirrel's heart rate drops from 200 beats per minute to about 5. Its body temperature drops from 98 degrees F to just above freezing. It breathes only once or twice per minute. This is a controlled near-death state that allows the animal to survive months without eating by burning almost zero energy.